

Homework #3

Due: 11:59pm, February 9

OS denotes the textbook Oppenheim & Schaffer “Discrete-Time Signal Processing”

1. [OS 2.32: Action of DT system on sinusoid]

Consider an LTI system with frequency response

$$H(e^{j\omega}) = e^{-j(\omega - \frac{\pi}{4})} \left(\frac{1 + e^{-2j\omega} + 4e^{-j4\omega}}{1 + \frac{1}{2}e^{-j2\omega}} \right), \quad -\pi < \omega \leq \pi.$$

Determine the output $y[n]$ for all n if the input for all n is

$$x[n] = \cos\left(\frac{\pi n}{2}\right).$$

2. [OS 2.37: Output from periodic input]Consider a discrete-time LTI system with impulse response $h[n]$. If the input $x[n]$ is a periodic sequence with period N (i.e. if $x[n] = x[n + N]$), show that the output $y[n]$ is also a periodic sequence with period N .**3. [OS 2.78: Time-domain manipulations]**Let $x[n]$ and $X(e^{j\omega})$ represent a sequence and its DTFT, respectively. Determine, in terms of $X(e^{j\omega})$, the transform of $y_s[n]$, $y_d[n]$, and $y_e[n]$. In each case, sketch $Y(e^{j\omega})$ for $X(e^{j\omega})$ given by

$$X(e^{j\omega}) = 1 - |\omega/\pi|, \quad -\pi < \omega \leq \pi.$$

(a) Sampler:

$$y_s[n] = \begin{cases} x[n] & n \text{ even,} \\ 0 & n \text{ odd.} \end{cases}$$

Note that $y_s[n] = \frac{1}{2}x[n] + \frac{(-1)^n}{2}x[n]$.

(b) Compressor:

$$y_d[n] = x[2n].$$

(c) Expander:

$$y_e[n] = \begin{cases} x[n/2] & n \text{ even,} \\ 0 & n \text{ odd.} \end{cases}$$

4. [OS 2.41: Filter]

- (a) Let $L \in \mathbb{N}$. Define

$$f(\omega) = \sum_{l=-\infty}^{\infty} \delta\left(\omega - \frac{2\pi l}{L}\right).$$

Plot $f(\omega)$.

- (b) Compute the Fourier series representation of $f(\omega)$. Determine the smallest $T > 0$ such that $f(\omega + T) = f(\omega)$. Then find the sequence $(a_k)_{k \in \mathbb{Z}}$ such that

$$f(\omega) = \sum_{k=-\infty}^{\infty} a_k e^{j2\pi k\omega/T}.$$

Plot the sequence $(a_k)_{k \in \mathbb{Z}}$.

- (c) Define

$$x[n] = \frac{L}{2\pi} \sum_{m=-\infty}^{\infty} \delta[n + Lm].$$

Then plot $x[n]$ and verify that

$$a_k = x[-kL].$$

- (d) From the above results, deduce that the DTFT of $x[n]$ is given by

$$X(e^{j\omega}) = f(\omega).$$

- (e) An LTI system has frequency response

$$H(e^{j\omega}) = \begin{cases} e^{-j3\omega}, & |\omega| < \frac{2\pi}{16} \left(\frac{3}{2}\right), \\ 0, & \frac{2\pi}{16} \left(\frac{3}{2}\right) \leq |\omega| \leq \pi. \end{cases}$$

Find the output $y[n]$ of the system from the input $x[n]$ when $L = 16$.

5. [Multichannel blind deconvolution]

Consider two causal LTI systems with impulse response $h_1[n]$ and $h_2[n]$. Both responses have a finite support of length 3 (i.e., $h_k[n] = 0$ for $n < 0$ and $n \geq 3$). These systems are driven by a common causal input $x[n]$ with finite support of length N (i.e., $x[n] = 0$ for $n < 0$ and $n \geq N$). The outputs are given by $y_k[n] = h_k[n] * x[n]$ for $k = 1, 2$.

Let $X(z)$, $H_k(z)$, and $Y_k(z)$ denote the z -transforms of the input, impulse responses, and outputs, respectively.

- (a) **Polynomial Analysis:** Show that $X(z)$, $H_k(z)$, and $Y_k(z)$ are polynomials in z^{-1} . Determine the degree of each polynomial in terms of N .
- (b) **Scaling Ambiguity:** Let α be a nonzero constant. Define scaled signal $\hat{x}[n] = \alpha x[n]$ and $\hat{h}_k[n] = \alpha^{-1} h_k[n]$.
- Show that the outputs generated by these scaled signals, $\hat{y}_k[n]$, are identical to the original outputs $y_k[n]$.

- Conclude that $x[n]$ and $h_k[n]$ can be only be determined from the outputs up to a constant scalar factor.
- (c) **Root Identification:** Assume that $H_1(z)$ and $H_2(z)$ share no common roots (they are coprime). Show that the roots of $X(z)$, $H_1(z)$, and $H_2(z)$ can be uniquely identified from the roots of the output polynomials $Y_1(z)$ and $Y_2(z)$. **Hint:** Recall that by the Fundamental Theorem of Algebra, any polynomial of degree M , $p(v) = \sum_{k=0}^M a_k v^k$ (with $a_M \neq 0$), can be uniquely factored as $p(v) = a_M \prod_{k=1}^M (v - r_k)$.